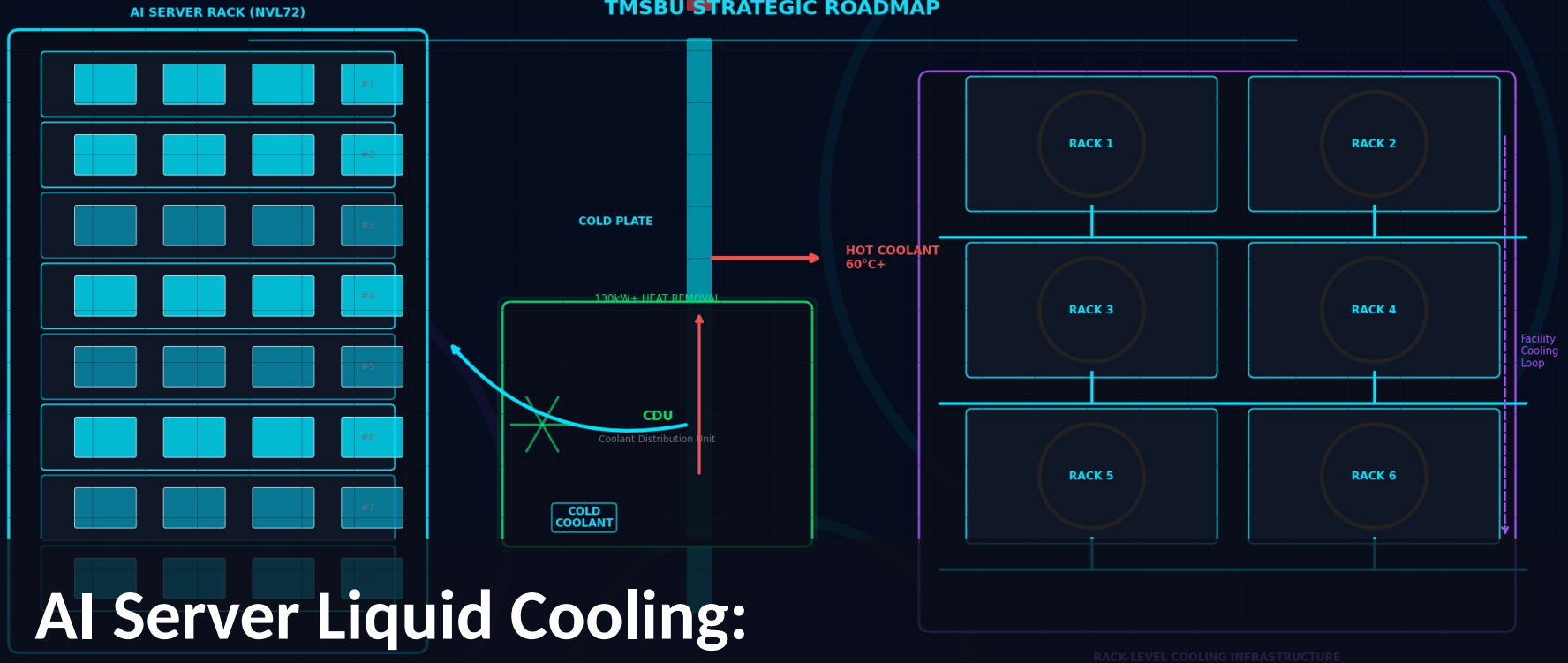


AI SERVER LIQUID COOLING

TMSBU STRATEGIC ROADMAP



AI Server Liquid Cooling:

TMSBU Commercial Opportunity & Strategic Roadmap

V7 — Final Content & Executive Polish

Executive Summary

02

Liquid cooling is not only a product opportunity — it is a business model transition for TMSBU.

01



1,000W+
Next-gen GPU thermal load

AI rack power density is moving toward 100kW+ class, pushing traditional air-cooling architecture toward practical limits.

02



60%*
Liquid cooling adoption by 2026

Liquid cooling adoption is accelerating under analyst forecasts; 2026 adoption rate requires source validation.

* Analyst forecast — 2026 adoption rate requires source validation

03



AIR + LIQUID
Hybrid architecture becomes standard

GPU 本體用液冷，memory、PCB、power 仍需風冷。兩者長期共存。

04



\$0.8B-\$3.2B
Scenario-based TMSBU opportunity

建立液冷系統能力、切入 NVIDIA 參考設計供應商名單、與 Tier 1 客戶建立 Thermal TI。

NOW

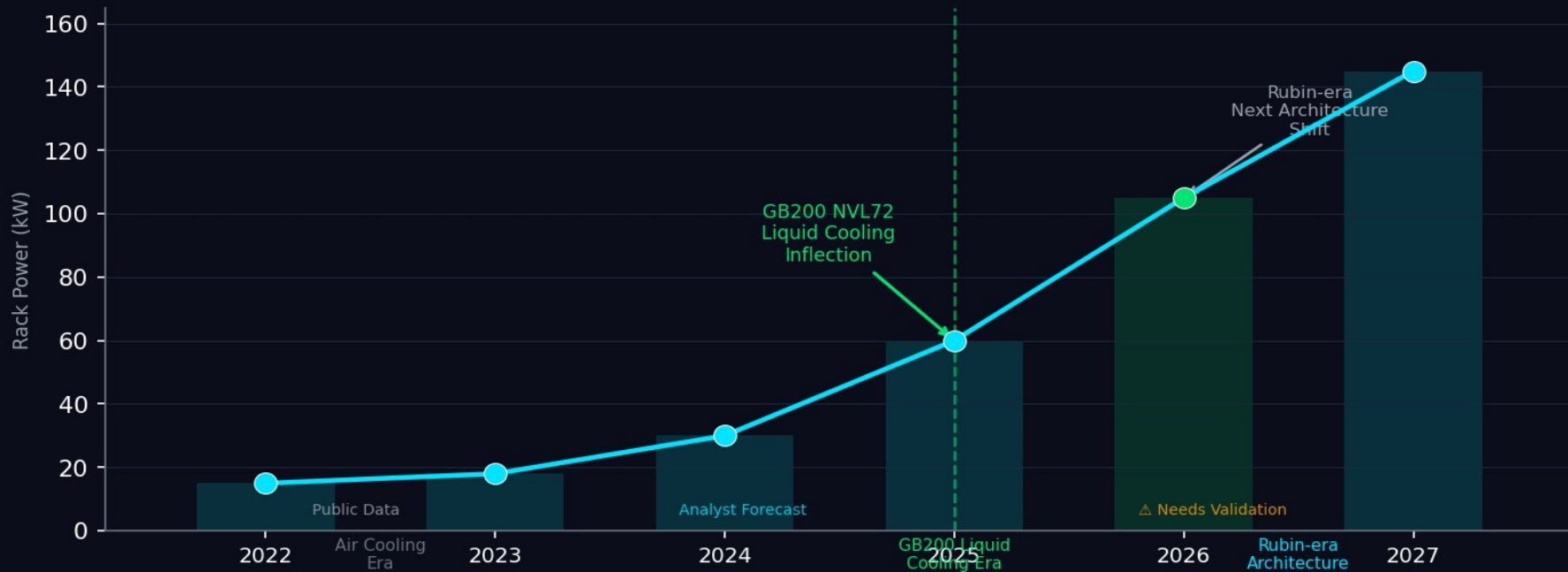
05
6-18M design-in window

建立液冷系統能力、切入 NVIDIA 參考設計供應商名單、與 Tier 1 客戶建立 Thermal TI。

Market Inflection: AI Compute Density Drives Thermal Architecture Shift

03

液冷在 2025 年 GB200 NVL72 達到爆發點，2027+ Rubin 架構將功率密度推向更高



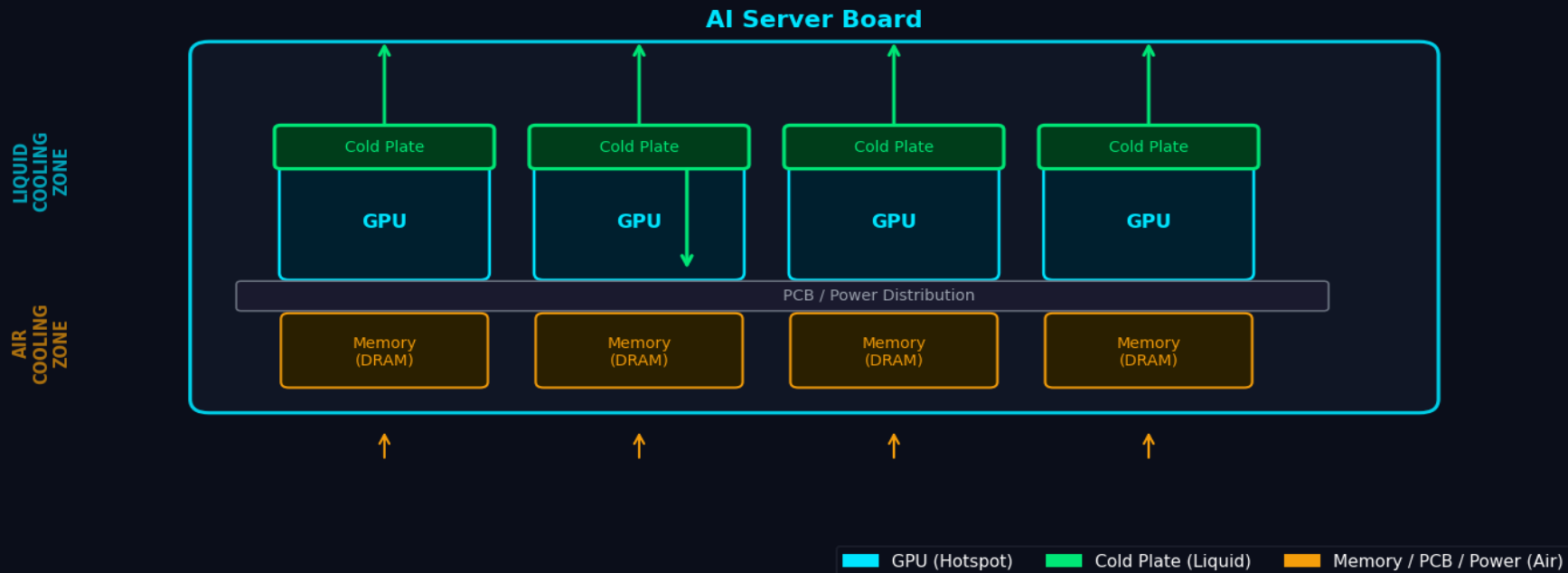
KEY INSIGHTS

- AI rack power density is moving toward 100kW+ class, pushing traditional air-cooling architecture toward practical limits.
- 80kW+ rack power class is becoming increasingly relevant for next-gen AI infrastructure; adoption depends on platform and customer configuration.
- TMSBU existing air-cooling portfolio must evolve toward hybrid air + liquid modules to capture next-generation AI server demand.

Hybrid Air + Liquid: Hotspots vs. System-Wide Cooling

04

液冷處理 GPU 熱點，風冷處理其餘系統組件——兩者長期共存

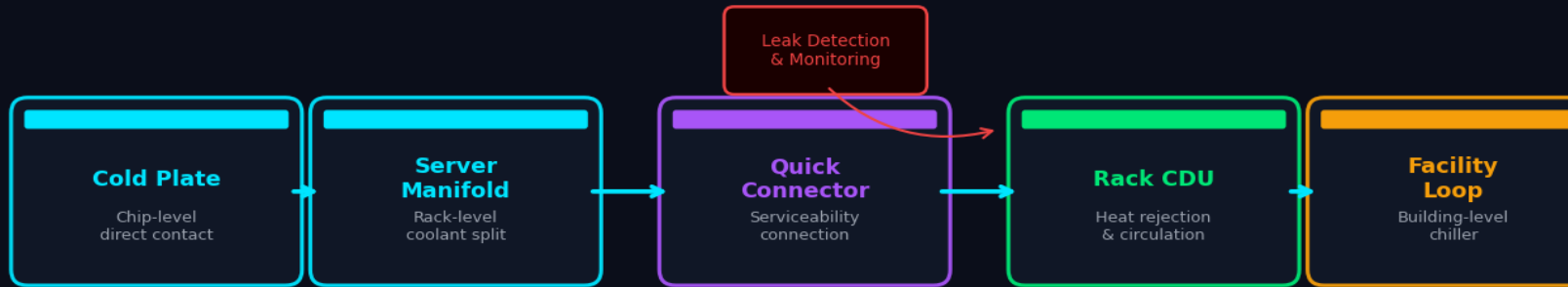


KEY INSIGHTS

- Cold plate enables direct-to-chip heat removal for high-power GPU / CPU hotspots.
- Hybrid architecture is likely to be a mainstream transition path for 2025–2028 AI server platforms.
- Air cooling remains necessary for many non-liquid-cooled components such as memory, PCB, power delivery, networking, and storage.

Chip-to-Rack Liquid Cooling Architecture

System value comes from integration: thermal, fluid, control, and serviceability must work as one architecture.



Source: Industry projection · Some specs need TMSBU internal validation

Coolant Flow →

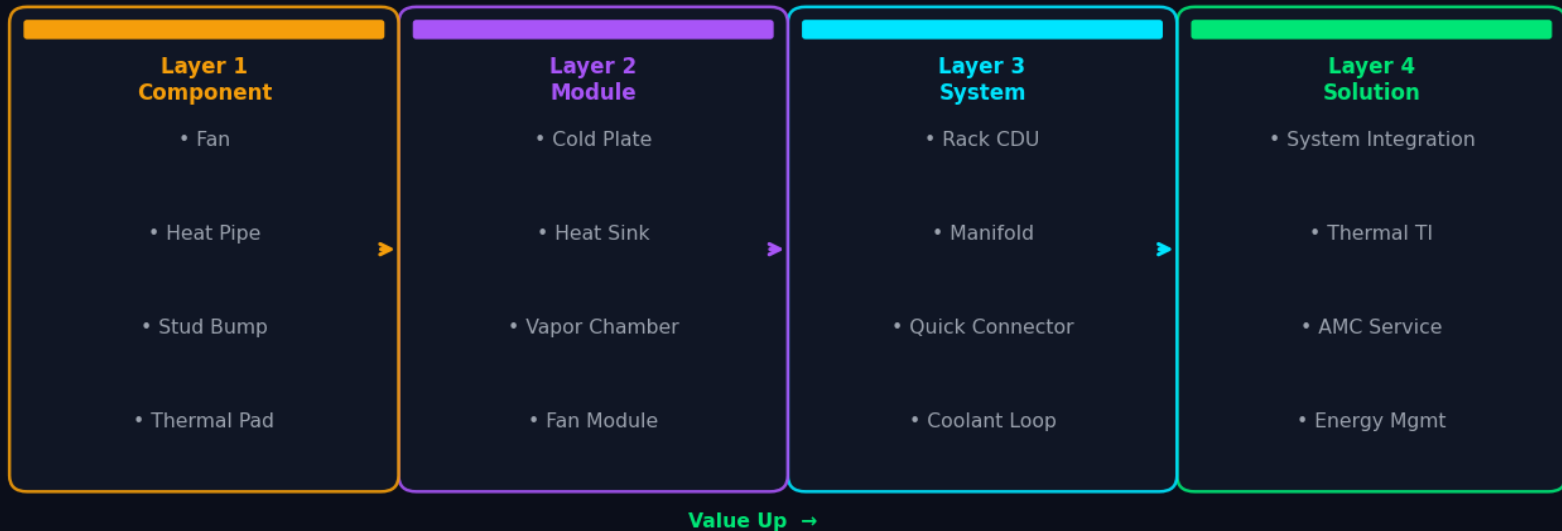
KEY INSIGHTS

- Quick Connector positioned at server-to-rack service interface, not as a main heat flow node.
- Leak detection and monitoring add system-level reliability for mission-critical AI deployments.
- CDU capacity Δ requires TMSBU internal validation — no specific kW figure used in this deck.

Architecture based on industry-standard direct-to-chip cooling loop. CDU specs Δ require TMSBU internal validation.

TMSBU Portfolio: 4-Layer Value Upgrade Path

Component → Module → System → Solution : 每層疊加更高客戶價值



Delta Group Synergy: Four Quadrant Advantage

07

Power Electronics × Thermal Management × Control Systems × Infrastructure

Power Electronics

- Power supply
- UPS systems
- Energy conversion

Thermal Management

- Cold plates
- Heat sinks
- CDU / manifolds

Sensor / Monitoring / Control

- Thermal sensors
- Auto control
- BMS integration

Infrastructure

- Rack PDU
- Coolant distribution
- Facility integration

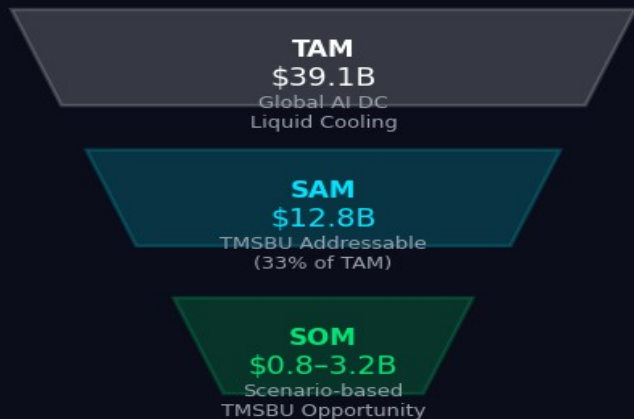
Customer Value: Lower integration risk · Faster design-in · Better reliability · Stronger energy-efficiency support

Market Opportunity: \$0.8B–\$3.2B Scenario-Based Opportunity

08

Scenario-based TAM/SAM/SOM framework — *⚠ SOM requires internal TMSBU pipeline validation*

Market Funnel



TMSBU Opportunity Scenarios



⚠ SOM is scenario-based and requires TMSBU internal validation

⚠ SOM is scenario-based and requires TMSBU internal validation

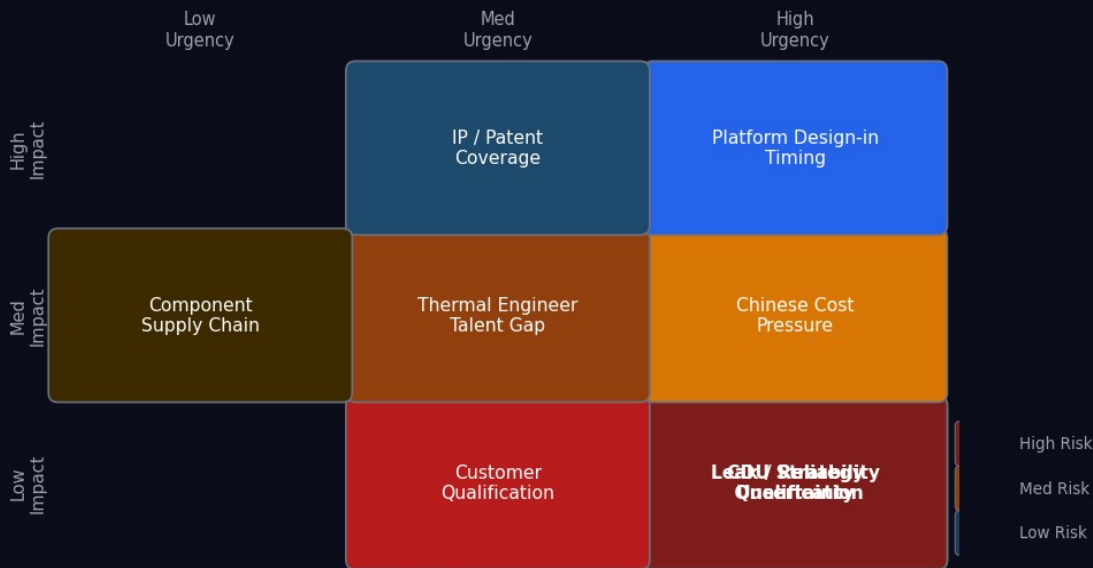
† Aggressive scenario assumes system-solution expansion and multiple tier-1 design wins

Capability Gap & Risk Assessment

09

Impact × Urgency matrix — immediate priorities need executive decision

Impact × Urgency Risk Matrix



TOP 3 IMMEDIATE PRIORITIES

1 CDU Strategy

Build · Partner · Acquire
decision required in 90 days

2 Leak Reliability

Validate quick connector,
leak detection, and
system-level reliability.

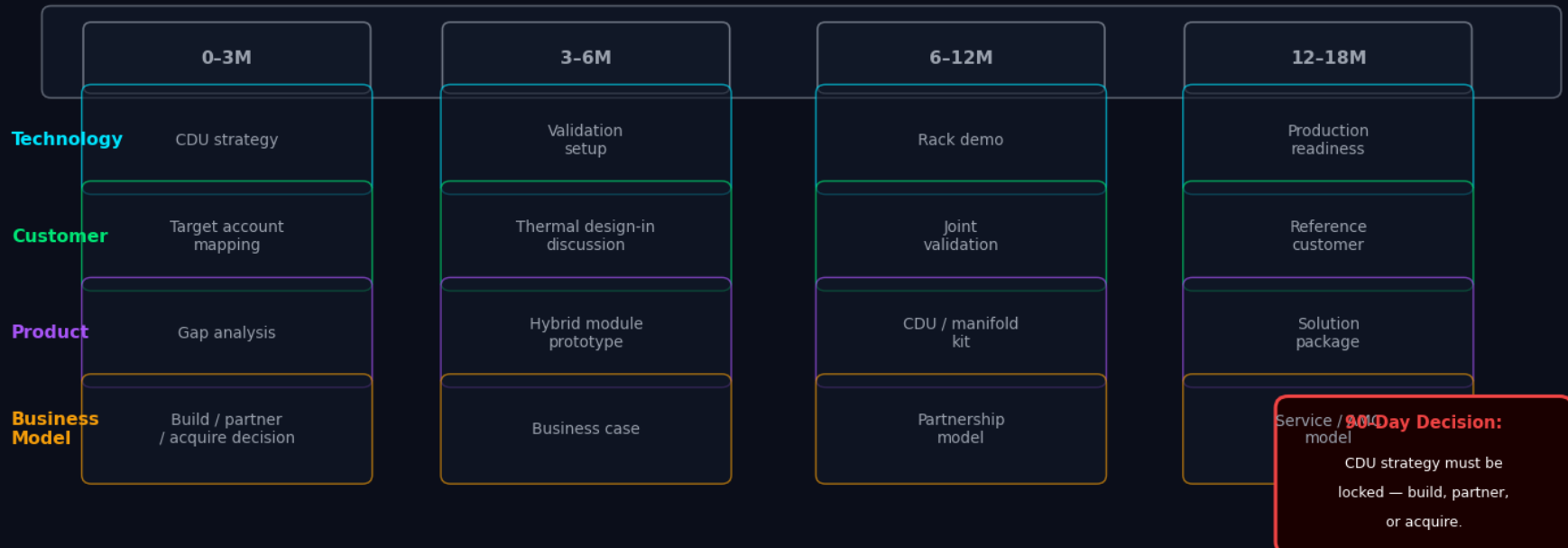
3 Customer Qualification

Start 12-18M design-in cycle
now to avoid missing
next-gen platform engagement.

Strategic Roadmap: 4-Lane Execution Plan

10

Technology · Customer · Product · Business Model — 6–18 month window



KEY INSIGHTS

- Phase 1 (0–6M) : 確認 CDU 策略，鎖定目標帳戶，展開 Thermal Design-in 討論
- Phase 2 (6–12M) : CDU 原型驗證，與 Tier-1 客戶建立 Joint Validation
- Phase 3 (12–18M) : CDU / Manifold / Quick Connector 產品化，通過可靠性認證